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Design and Evaluation of a Legal AI Assistant: A Comparative Study of RAG and Fine-Tuning

TER Final Report

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Abstract

This report compares two ways of adapting a large language model to a specialized domain, retrieval-augmented generation (RAG) and parameter-efficient fine-tuning (QLoRA). The chosen domain is law, a demanding field where a wrong answer or a fabricated citation has a real cost. We build a French-language legal question-answering assistant, evaluated on BSARD, a French-speaking Belgian law corpus, using the Mistral-7B-Instruct model. Four configurations are compared on 222 test questions, without adaptation, with RAG alone, with fine-tuning alone, and with both combined. Answer quality is measured through automatic metrics (ROUGE-L, BERTScore, citation exact match, hallucination rate), a complementary automatic judgment (LLM judge, fidelity), and a blind human evaluation. RAG and fine-tuning each bring a clear gain over no adaptation, and combining them does even better on average. But neither technique on its own solves the most serious problem observed, namely that the system confidently fabricates legal references rather than acknowledging its limits.

1. Introduction

A large language model can today write fluent text on almost any topic, but its knowledge remains that of its training data, general-purpose, often in English, and frozen at a given date. As soon as one wants to use it on a precise and demanding domain, the question arises of how to give it access to knowledge it does not natively possess. Two families of technical answers dominate the literature and practice. Fine-tuning partially retrains the model on domain data. Retrieval-augmented generation, or RAG, leaves the model untouched but supplies it, at answer time, with relevant document excerpts retrieved from an external knowledge base. These two families are not just two different implementations of the same objective. They do not make the same choice about where to store this new knowledge. Fine-tuning places it directly inside the model's weights, whereas RAG keeps it outside, in a knowledge base consulted at answer time. It is this difference in principle that justifies comparing the two rather than picking one out of habit. This difference is not purely academic. An external document base is updated simply by chang-

ing the indexed documents, whereas a fine-tuned model must be retrained as soon as the law changes, which carries a very different operational cost depending on the technique chosen.

This work experimentally compares these two approaches on a concrete use case, an assistant able to answer legal questions in French while citing the law article it relies on. Three research questions guide the whole work.

1. Can a model specialized through fine-tuning, even trained on few examples, rival a general-purpose model backed by external document retrieval ?
2. In which cases does fine-tuning bring a real gain over RAG alone, and does this gain hold once combined with RAG ?
3. Can the system reliably cite its source, and does it recognize its own limits when it cannot answer correctly ?

1.1 Why law

Choosing law as the experimental ground is not arbitrary. An approximate legal answer is not merely less useful than a precise one, it can actively mislead a person relying on it for a real decision, which gives the question of reliability concrete weight. Source traceability is also a requirement of the domain itself rather than an artificially added constraint. A credible legal answer rests on an identifiable piece of legislation, which gives a measurable reliability signal (does the cited reference really exist, and is it the right one) where many other domains only offer a subjective quality judgment. Finally, the hierarchical structure of legal text, code, book, title, chapter, article, naturally lends itself to splitting into citable units, which makes it a good use case for RAG.

Other candidate domains were ruled out for practical rather than theoretical reasons. The medical domain shares several of these properties, but access to real data there is heavily restricted for confidentiality reasons. The availability of freely accessible French-language legal corpora (section 3) settled the choice of domain even before the experiments began.

The rest of the report presents the project’s institutional context, reviews the state of the art that guided our choices, describes our methodological contribution, presents the full set of results obtained, and then concludes by answering the three questions above.

2. Project context

This work is a TER (*travail d’étude et de recherche*, a supervised student research project), carried out in pairs at Université Jean Monnet de Saint-Étienne.

The topic compares RAG and fine-tuning on a concrete use case. All GPU computation was carried out on Kaggle, using a Tesla T4 graphics card. This constraint directly shaped methodological choices described later, the size of the fine-tuning training sets and the choice of a 7-billion-parameter model.

3. State of the art

3.1 Efficient adaptation of large language models

Fully fine-tuning an LLM with several billion parameters is costly in memory and compute time, which motivated the development of parameter-efficient fine-tuning methods (PEFT). LoRA [10] proposes freezing the pre-trained model weights and adding, alongside certain matrices (typically the attention matrices), a pair of low-rank matrices whose product approximates the update that a full fine-tuning would have produced

$$W' = W_0 + BA,$$

with W_0 the original frozen matrix, B of size $d \times r$, A of size $r \times k$, and the rank r chosen to be much smaller than d or k . The number of trainable parameters thus becomes a tiny fraction of the total model. QLoRA [6] takes this idea further by quantizing the base model to 4 bits (NF4 format) before attaching the LoRA adapters, which sharply reduces the memory footprint and makes it possible to train a 7-billion-parameter model on a single consumer-grade GPU. The quantized weights remain frozen throughout training, only the adapters are updated, which avoids having to store gradients and an optimizer state for the entire model. This is the approach we use (section 4.4).

3.2 Retrieval-augmented generation

RAG couples a document retrieval model with a generative model. At answer time, the most relevant passages from a document base are retrieved and then injected into the context given to the generator, which avoids retraining the model each time the base is updated and, in principle, allows the source used to be cited explicitly. Document retrieval relies either on dense representations obtained from an embedding model [12], on lexical methods such as BM25, or on a combination of the two. BM25 scores a document based on the frequency of the question’s words it contains, weighted by their rarity across the whole corpus, which makes it very effective

on exact terms but blind to synonyms. The two signals can be combined through reciprocal rank fusion, which assigns each document a score

$$\text{RRF}(d) = \sum_i \frac{1}{k + \text{rang}_i(d)},$$

the sum running over the rankings produced by each method, and k a constant that reduces the weight of the lowest ranks, so that a document ranked well by at least one method rises in the final ranking even if it is absent from the other. Two dense texts are compared through the cosine similarity between their vectors

$$\text{sim}(q, d) = \frac{q \cdot d}{\|q\| \|d\|},$$

which brings semantically close texts together even when they share few exact words, a property a purely lexical method does not have. Our own pipeline, a so-called naive RAG rather than an advanced variant, is described in section 4.3.

3.3 Existing comparisons between RAG and fine-tuning

Our work positions itself directly relative to the Derby LLM study by Bouvard, Ciancone, Gourru, and Schaeffer [2]. This study compares RAG and fine-tuning, using the same base model as us, Mistral 7B. Their fine-tuning proceeds by continued pre-training, that is a continuation of the next-word prediction task directly on the domain documents, without reformulation into question-answer pairs, whereas their RAG relies on a commercial solution and on OpenAI embeddings. Their main conclusion is clear, RAG is very strongly preferred by human annotators (around 67 % versus 3 % for fine-tuning depending on the dataset), and fine-tuning by continued pre-training struggles to reliably inject new knowledge.

Our protocol deliberately reuses the same families of metrics (pairwise arena, entity-overlap fidelity) to allow a direct comparison, but differs on one important point, our fine-tuning is supervised on question-answer pairs formatted with an article citation, rather than unsupervised continued pre-training. This distinction connects to the work of Ovadia et al. [18], who show that RAG systematically outperforms unsupervised fine-tuning for injecting factual knowledge, exactly the limitation encountered with fine-tuning by continued pre-training. By testing a supervised fine-tuning, closer to the use case QLoRA was designed for, we check whether this limitation still holds.

3.4 Evaluating a generative question-answering system

Evaluating text generated by an LLM remains an open problem, and several families of methods coexist, each with its own limitations. Surface similarity metrics such as ROUGE [14] count the overlap of word subsequences between a generated text and a reference. They are fast

and reproducible but unfairly penalize a correct rephrasing that does not use the same words. BERTScore [20] partially fixes this flaw by comparing contextual representations rather than exact strings, at the cost of more expensive computation. More recent tools such as RAGAS [7] specifically evaluate RAG systems by breaking quality down into several facets, typically by having an LLM extract and then verify individual claims.

Human evaluation through direct pairwise comparison of answers gives a signal closer to real-world use but remains costly to collect and subject to imperfect inter-annotator agreement, which must itself be measured rather than assumed. The Bradley-Terry model, associated with this method, turns a set of pairwise votes into a relative strength score on a single scale, and we reuse it for our own arena (section 5.7). Finally, using a large language model as a judge introduces documented biases, in particular a self-preference bias when the judge is also one of the evaluated systems, a point we took care to avoid in our protocol by choosing a judge external to the compared configurations (section 5.7).

3.5 Corpora, benchmarks, and base models

Several legal benchmarks exist in the literature, mostly in English. LegalBench [8] gathers a collaboratively built collection of legal reasoning tasks, CUAD [9] focuses on contract analysis, and LexGLUE [3] proposes a set of legal language understanding tasks. For French, BSARD [15] provides a French-speaking Belgian statutory article retrieval corpus, described in detail in section 4.1. Its natural extension, LLeQA [16], adds written answers on this same corpus, and was our initial target, but access was never granted despite a request, which justified the switch to BSARD. This switch has a direct consequence on evaluation, discussed in section 4.1, a document retrieval corpus does not provide a written reference answer, unlike what LLeQA would have offered, which forced us to reconstruct this reference ourselves rather than being able to rely on an already existing human-written answer.

On the model side, we use Mistral-7B-Instruct [11], a 7-billion-parameter dense decoder-only transformer released as an open-weight model by Mistral AI, chosen for its good trade-off between quality and running cost and for a level of French-language competence that most large language models, trained predominantly on English text, do not match. We also test whether our observations generalize to Qwen2.5-7B-Instruct (section 5.3), a model of comparable size released by Alibaba but trained on a different, larger pretraining mixture, which makes it an independent comparison point and not merely a variant of the same model. All these models are based on the Transformer architecture [19] and are aligned through instruction tuning following InstructGPT [17], a step that trains them to follow an instruction rather than simply continue a text, and which explains why they can be queried directly with a question without prior examples.

4. Methodological contribution

4.1 Data

BSARD (Belgian Statutory Article Retrieval Dataset) [15] is a public French-speaking Belgian law corpus, made up of 22 633 law articles and questions annotated with the identifiers of the articles needed to answer them. The corpus provides an 886-question training split and a 222-question test split, which we use in full rather than on a subsample, precisely so that our conclusions do not depend on a particular sampling draw. Each question carries a legal category field with seven possible values (figure 1), whose distribution is strongly imbalanced, Family and Housing dominate by far while Social protection has only four questions in the test set, a limitation to keep in mind when reading the results by category (section 5.5).

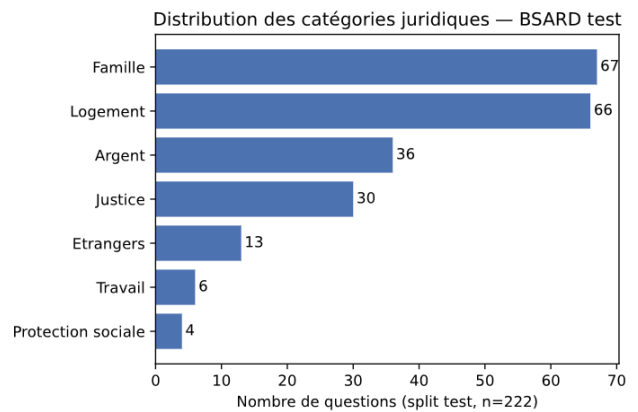


Figure 1. Distribution of BSARD test questions by legal category.

The choice of French-speaking Belgian law rather than French law deserves justification. The object of this work is comparing methods for adapting a model to a domain, not French legal content as such. The language remains French, the structure of the problem (citing a precise law article to answer) is identical, and the blocked access to LLeQA makes BSARD the best freely available substitute.

A methodological limitation must be explicitly acknowledged. BSARD is natively a document retrieval corpus rather than a generative question-answering one, it contains no natural-language written answer, only the identifiers of the relevant articles. The reference answer used to compute ROUGE-L and BERTScore is therefore reconstructed by concatenating the text of the cited articles, which is not an ideal answer such as a legal professional would have written. The surface similarity scores presented later should therefore be read as a measure of lexical overlap with the statutory text, not as a measure of writing quality. This limitation largely explains why all the ROUGE-L scores observed in this report remain relatively low, without this meaning that the answers are unusable.

4.2 Four compared configurations

We compare four configurations, summarized in table 1. The same system prompt, in French, is used in all four cases to ensure a fair comparison. It asks for a clear, written answer, grounded in the provided context where applicable, mandatorily ending with a citation in the format “Article <number>”, and explicitly invites the model to answer “I don’t know” rather than inventing an answer if it does not have enough information.

Table 1. Description of the four compared configurations.

Config	Retrieval	Fine-tuning
C1 zero-shot	no	no
C2 RAG	yes	no
C3 fine-tune	no	yes
C4 fine-tune + RAG	yes	yes

4.3 Retrieval pipeline

Document retrieval combines a dense index, built with *multilingual-e5-large*, a multilingual sentence-embedding model released by Microsoft and built on the XLM-RoBERTa-large backbone, trained to map text in over 100 languages into a shared vector space so that semantically similar sentences end up close together regardless of exact wording, and a lexical BM25 index, merged through reciprocal rank fusion. A cross-encoder reranker, *bge-reranker-v2-m3* [4], part of the BGE family of multilingual retrieval models released by BAAI, then reorders the retrieved candidates, scoring each question-article pair jointly instead of comparing independently computed vectors, which is more accurate but too slow to run on the whole corpus. By default, the five top-ranked articles are injected into the generator’s prompt. This two-stage pipeline choice (retrieval then reranking) comes directly from the state of the art (section 3.2), but its actual contribution on our corpus was not a given in advance, which justifies the dedicated ablation in section 5.1. Figure 2 summarizes the pipeline end to end.

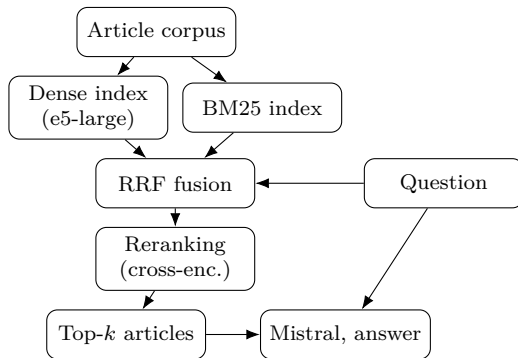


Figure 2. Retrieval pipeline. The corpus is indexed once through parallel dense and lexical routes, merged by reciprocal rank fusion and reordered by a cross-encoder reranker. The question drives the search and is injected into the final prompt with the top- k articles.

4.4 QLoRA fine-tuning

A full fine-tuning of Mistral-7B in standard precision would require storing the model, its gradients, and the optimizer state, that is far more memory than a consumer-grade GPU offers. QLoRA gets around this wall by keeping the base model in 4 bits throughout training and computing gradients only for the small matrices B and A added by LoRA, which makes training on a T4 possible. The 4-bit format used, NF4, is designed around how pretrained network weights are actually distributed, most values clustered near zero and large values rare, rather than treating every value as equally likely the way a plain integer format would, and QLoRA further compresses the quantization constants themselves through a second, lighter quantization pass. Restricting training to the adapter matrices also keeps the optimizer state small, since it only needs to be tracked for parameters that are actually updated, which is the main memory cost that 4-bit weights alone do not remove.

Concretely, fine-tuning freezes the base model, quantized to 4 bits, and trains a rank-32 LoRA adapter on the attention matrices, that is around 0.375 % of the model’s total parameters. This rank was not chosen arbitrarily, it was selected after comparison with a lower rank (section 5.6), which gives a net but modest gain for a markedly higher compute cost. The main training run uses 580 question-answer-citation examples built from the BSARD training split, over three epochs, with a learning rate of 2×10^{-4} . One hundred additional questions, drawn with a fixed seed independent of the training seed, are held out for validation, identically regardless of the variant being tested. This fixed validation set is what makes the various rank, target, and training-size ablations presented in section 5.6 comparable with one another, rather than only tracking the training loss, which says nothing about the model’s ability to generalize to never-seen questions. Figure 3 summarizes this pipeline end to end, contrasting it with figure 2, no retrieval step is involved once the adapter is trained, and no document ever enters the prompt.

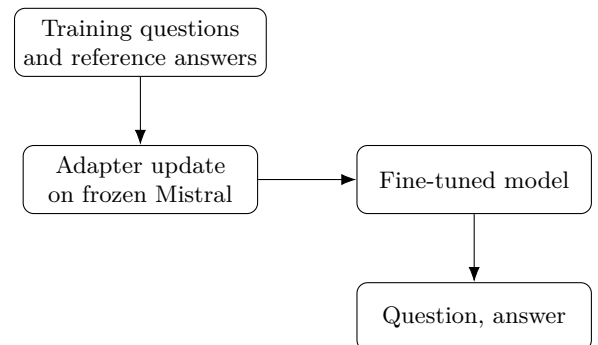


Figure 3. Fine-tuning pipeline. Only the small adapter is trained, the base weights never change, and no retrieval step is involved at inference time.

4.5 Statistical protocol

Unlike a small-sample comparison with a single number per configuration, each configuration is evaluated on the full set of 222 test questions, and each mean score is accompanied by a 95 % confidence interval obtained through bootstrap resampling. Concretely, this technique consists of randomly drawing, with replacement, 222 questions out of the 222 available, some therefore appearing several times in a given draw and others not at all, recomputing the mean score on that draw, and repeating the operation 1000 times. The interval bounds are then read directly from this simulated distribution of 1000 scores, by discarding the lowest 2.5 % of values and the highest 2.5 % of values, which gives a range of plausible values rather than a single value treated as exact.

For each pair of configurations, two significance tests are applied in parallel. The paired bootstrap test reuses the same resampling principle, but applied this time to the question-by-question score differences between the two compared configurations. It counts, across the 1000 draws, in what proportion this mean difference changes sign relative to the difference observed on the real data, and this proportion serves as a p-value estimate, without assuming any particular shape for the distribution of differences. The Wilcoxon signed-rank test proceeds differently, it ranks the differences by increasing absolute value, separately sums the ranks of the positive and negative differences, and then compares the smaller of the two sums to its theoretical distribution under the hypothesis of no difference. When the two tests agree, the conclusion is treated as solid. When they diverge, the comparison is treated as undecided rather than resolved in favor of the more favorable test. A Holm-Bonferroni correction is applied to account for testing six pairs at once, a so-called multiple comparisons problem, where testing several hypotheses in parallel mechanically increases the risk of finding a significant result by pure chance if no correction is applied. Concretely, the six p-values are ordered from smallest to largest, and the i -th smallest is compared to $\alpha/(6 - i + 1)$ rather than to a fixed threshold, an increasingly permissive threshold as one moves down the ranking, which controls the overall risk of false discovery across the six tests as a whole rather than on each one separately.

For the human evaluation (section 5.7), agreement between two annotators is measured by Cohen’s Kappa

$$\kappa = \frac{p_o - p_e}{1 - p_e},$$

with p_o the observed proportion of agreement, that is the share of questions where the two annotators gave exactly the same answer, and p_e the proportion of agreement expected by pure chance, computed from the frequency with which each annotator chooses each option independently of the other. Kappa equals 0 when the observed agreement is no better than that chance level, and 1 for perfect agreement, which avoids confusing high agreement with a simple shared bias toward a majority answer, two annotators who both very often favor the same con-

figuration by default can appear to agree without having actually judged each case on its own merits.

5. Results

5.1 Retrieval, evaluation, and improvement directions

Before turning to text generation, one thing had to be checked first, does document retrieval actually find the right law article. This point conditions everything else, no writing quality can make up for a missing article. We compared four retrieval methods rather than a single one, so as not to assume that a general-purpose embedding model would automatically be the best choice on a vocabulary as specialized as legal vocabulary. Table 2 presents these four methods on the 222 test questions, measured through three complementary indicators. Recall@1 measures whether the correct article ranks first. Recall@10 measures whether it appears anywhere among the top ten results. MRR@10 additionally accounts for its exact position, an article found in second position counts more than one found in tenth position.

Table 2. Comparison of retrieval methods on the 222 test questions.

Method	R@1	R@10	MRR@10
mpnet (initial)	0,021	0,097	0,105
BM25	0,063	0,224	0,215
e5-large	0,083	0,296	0,288
Hybrid (BM25+e5)	0,106	0,322	0,312

Moving from a general-purpose embedding model (mpnet) to a larger multilingual model (e5-large) noticeably improves Recall@1. More surprisingly, BM25, a purely lexical method that simply counts words shared between the question and the article with no learning at all, gets a better score than the initial general-purpose embedding model. This result suggests that legal vocabulary is poorly covered by embeddings trained on general-purpose text. Combining the two methods further improves every score. But even with this combination, the correct article is still missing more than two times out of three among the top ten results.

This ceiling pushed us to test three improvement directions rather than too quickly concluding an intrinsic difficulty of the corpus. Each of these directions targets a different explanation for the same problem, summarized in table 3

Table 3. Retrieval ablation, all variants on e5-large.

Variant	R@1	R@10	MRR@10
Base	0,083	0,296	0,288
+ Metadata	0,119	0,363	0,334
Chunking 256	0,083	0,330	0,306
+ Reranker	0,151	0,353	0,396

The first direction enriches each indexed fragment

with its hierarchical metadata, the code, the chapter, and the section it belongs to. It raises Recall@1 from 0.083 to 0.119, confirming that part of the initial difficulty came from a lack of textual cues rather than a hard limitation of the corpus. The second direction splits the articles into 256-token chunks, which yields 27 540 fragments versus 22 633 articles originally. It brings no net gain on Recall@1, however. This result is consistent with the length of BSARD articles, half of which are under 77 words, so the large majority simply does not need to be chunked. The third direction, a cross-encoder reranker, remains the most effective, with a Recall@1 of 0.151, nearly double the base score. It is also the least costly in compute time, since it only reworks a small number of already-retrieved candidates instead of searching the whole corpus (figure 4). This last point makes it a natural candidate to integrate by default into the pipeline rather than keep as a separate tested variant.

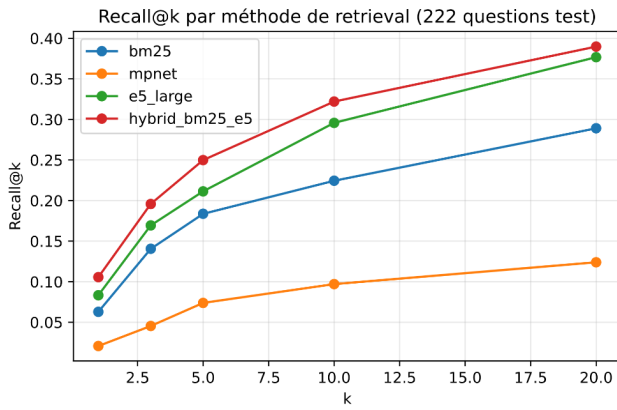


Figure 4. Recall@k by retrieval method, 222 test questions.

5.2 Overall comparison of the four configurations

Table 4 presents the main comparison, measured by ROUGE-L, a score that compares the generated text to the expected reference text and ranges from 0 for no resemblance to 1 for an identical text. Adding RAG alone or fine-tuning alone each markedly improves quality over no adaptation, and combining the two does even better on average. Each mean score is accompanied by a 95 % confidence interval (the CI low and CI high columns), obtained through bootstrap resampling.

Table 4. Mean ROUGE-L with 95 % bootstrap confidence intervals.

Config.	ROUGE-L	CI low	CI high
C1 zero-shot	0,110	0,104	0,116
C2 RAG	0,140	0,129	0,151
C3 fine-tune	0,147	0,136	0,160
C4 fine-tune+RAG	0,179	0,157	0,202

The paired significance tests, however, qualify this raw ranking, a higher score on 222 questions does not

by itself prove that a method is truly better, the gap could stem from the chance of drawing these particular questions. Two different tests were therefore applied to each pair of configurations. The paired bootstrap test directly compares, question by question, the score gap between two configurations. The Wilcoxon signed-rank test, more conservative, does not look at the size of this gap but only at whether one method wins more often than the other. Agreement between the two tests is read as a solid conclusion, while a disagreement leaves the comparison open rather than settled by whichever test happens to look more favorable. The difference between C2 and C3 is not significant, both tests agreeing on this, which means RAG and fine-tuning are statistically indistinguishable from each other on ROUGE-L despite what the raw means suggest. The comparison between C3 and C4 is the only case where the two tests disagree (the bootstrap judges it significant, Wilcoxon does not) and is therefore treated here as undecided.

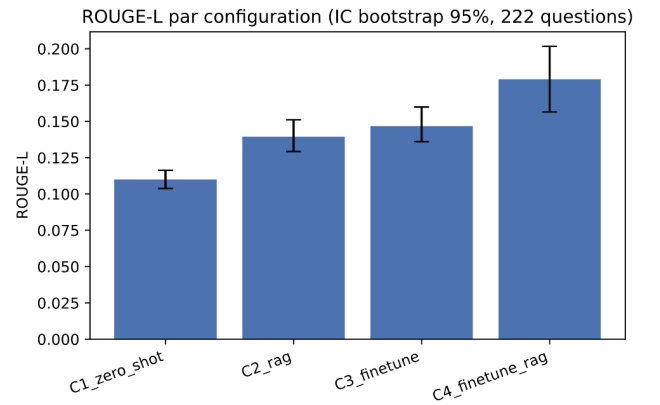


Figure 5. ROUGE-L with confidence intervals for the four configurations.

5.3 Citation reliability, error origin, and abstention

Since article citation is at the heart of the topic, table 5 looks at this figure separately from the overall lexical overlap measured by ROUGE-L.

Table 5. Answer reliability, citation exact match, and hallucination.

Config	Precision	Recall	Exact	Halluc.
C1	0,016	0,011	0,005	0,144
C2	0,176	0,132	0,036	0,234
C3	0,023	0,012	0,000	0,050
C4	0,069	0,053	0,005	0,167

Four indicators are used. Precision is the share of cited articles that are actually correct. Recall is the share of expected gold articles that the model actually cited. Exact match requires precision and recall to both be perfect at once. The hallucination rate is the share of answers citing an article number that does not exist anywhere in the corpus.

RAG achieves by far the best citation exact match, which makes sense since it has access to the article’s real text at answer time. Fine-tuning alone, however, achieves the lowest hallucination rate of the four configurations, almost three times lower than RAG. Fine-tuning thus learns to be cautious, it fabricates fewer references, but without citing the correct one often either, whereas RAG cites correctly more often but also hallucinates more often when it gets it wrong.

To understand where RAG’s remaining error comes from, an oracle-context experiment was run, in which the true relevant articles are placed directly in the prompt without going through the search engine, which entirely removes retrieval error. The gap between real retrieval and oracle context is very clear on citation exact match (0.036 versus 0.311, a gap of 0.275), markedly larger than the gap between real retrieval and no context at all. Most of what is missing therefore comes from document retrieval failing to find the right article, rather than from the model failing to make use of a good article once it has it.

The hallucination rate is higher with real retrieval (0.234) than with no context at all (0.144), and drops again under oracle context (0.117). A plausible but wrong article seems to invite the model into more confident fabrication than a complete absence of information. This same pattern, RAG improves overall quality while also increasing hallucination risk, was found again by reproducing the C1/C2 protocol with Qwen2.5-7B-Instruct instead of Mistral, which rules out the hypothesis of a mere model-specific quirk.

Finally, none of the previous measures say whether the system can recognize that it should not answer. A set of 50 questions outside the system’s scope was built for this purpose, mixing questions with no legal content, questions about foreign law, and questions referencing an article removed from the index. The result is unambiguous, the correct abstention rate stays close to zero for all tested configurations. Even for a question entirely unrelated to law, the system produces a confident-looking answer, fabricated article numbers included, in nearly all cases. This is a clearest weakness, and neither of the two tested techniques corrects it on its own.

5.4 Two robustness checks

Before interpreting the citation gaps any further, two robustness points were checked.

The first concerns the format of article numbers. About 32 % of BSARD identifiers are not purely numeric (Belgian regional formats such as “L1122-9” or “R.II.21-7”), which raises the question of whether the model cites these irregular references less well, being harder to reproduce word for word while generating text. A Mann-Whitney test, which compares two groups of questions without assuming any particular distribution, was applied separately to each configuration between questions whose gold article has a simple identifier and those with a structured identifier.

Table 6. Citation exact match by gold identifier format, Mann-Whitney test.

Config	Simple	Structured	Chance
C1 zero-shot	0,85 %	0 %	35 %
C2 rag	6,78 %	2,88 %	18 %
C3 finetune	0 %	0 %	100 %
C4 finetune+rag	0,85 %	0 %	35 %

The difference does go in the expected direction wherever it is measurable, especially for RAG, but none reaches the usual 5 % significance threshold. This result mainly shows that there is not enough evidence to confirm an effect, rather than proving there is none, since citation exact match is a rare event, the test has very little material to decide with, particularly for fine-tuning alone, which has no successes in either group.

The second point concerns the stability of the measurements themselves. The model generates with a non-zero temperature, so two runs of the same configuration on the same questions can produce different answers. RAG was fully rerun three times on the 222 questions with three distinct random draws. ROUGE-L varies by less than 2 % relative from one run to another, but the rate of exactly correct citations varies by nearly 31 % relative, which concretely represents two to three questions out of 222 flipping from one run to the next. Since this rate is already low in absolute terms, this flip weighs heavily in proportion. This is why the significance tests in section 5.2 focus on ROUGE-L, the stable measure, rather than on citation exact match, more informative but noisier at non-zero temperature.

5.5 What makes a question difficult

The average performance hides gaps between legal categories and between questions. Figure 6 and table 7 break ROUGE-L down by category and by configuration.

Table 7. Mean ROUGE-L by legal category and by configuration.

Category	<i>n</i>	C1	C2	C3	C4
Family	67	0,117	0,146	0,179	0,182
Housing	66	0,118	0,140	0,149	0,216
Money	36	0,112	0,167	0,128	0,191
Justice	30	0,082	0,099	0,105	0,116
Foreign nationals	13	0,096	0,118	0,126	0,117
Labor	6	0,121	0,153	0,154	0,138
Soc. protection	4	0,087	0,126	0,112	0,137

The overall ranking of the four configurations broadly holds across categories, but with very different magnitudes. Housing shows the largest gap between C1 and C4 (0.118 to 0.216), suggesting that fine-tuning and RAG reinforce each other particularly well there, whereas on Labor fine-tuning alone already matches or exceeds the combined configuration. Social protection, with only four test questions, remains inconclusive and serves more as an illustration than as proof.

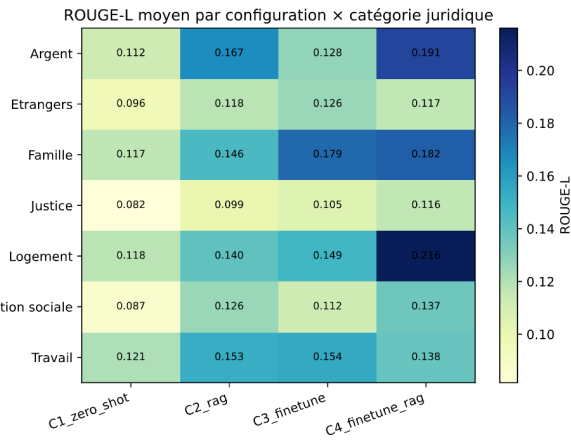


Figure 6. Mean ROUGE-L by configuration and legal category.

Beyond category, two hypotheses were tested directly to explain why some questions get better answers than others, question length and the number of law articles needed to answer, by computing a Spearman correlation with the resulting ROUGE-L for each. This correlation replaces raw values with their rank before computing the usual coefficient, which makes it sensitive to a monotonic relationship rather than specifically a linear one.

Table 8. Spearman correlation between difficulty and ROUGE-L.

Config	ρ length	ρ num. articles
C1	-0,107	-0,491
C2	-0,076	-0,394
C3	-0,064	-0,510
C4	-0,152	-0,353

Question length has almost no relation to answer quality. The number of required articles, however, correlates negatively and highly significantly with quality in all four configurations without exception ($p < 0,0001$ everywhere), which means length was the wrong difficulty indicator all along, the real factor being the number of distinct rules to combine in a single answer. Quality drops sharply as soon as four or more articles are needed, in every configuration, and neither RAG nor fine-tuning fixes this structural problem on its own.

This observation motivated one last question. Can we predict, from features observable before generation even happens (question length, number of gold articles, category, identifier format regularity, configuration), whether an answer is going to be unreliable? A logistic regression classifier and a Random Forest, trained on the 888 available observations with 5-fold cross-validation, do markedly better than a naive classifier (AUC of 0.878 and 0.838 versus 0.50 for the majority class, figure 7), a result to be qualified, however, by the marked imbalance of the target, only 1.1 % of answers achieve an exactly correct citation, which leaves this positive class very few examples to rely on.

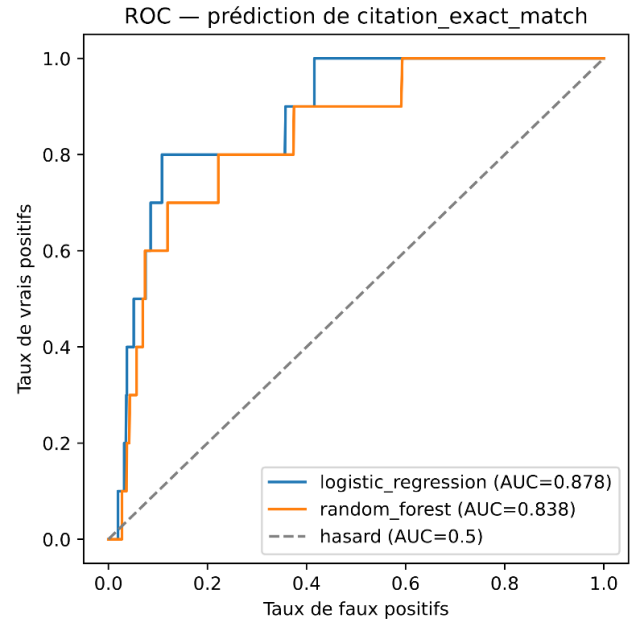


Figure 7. ROC curve of the two citation reliability classifiers.

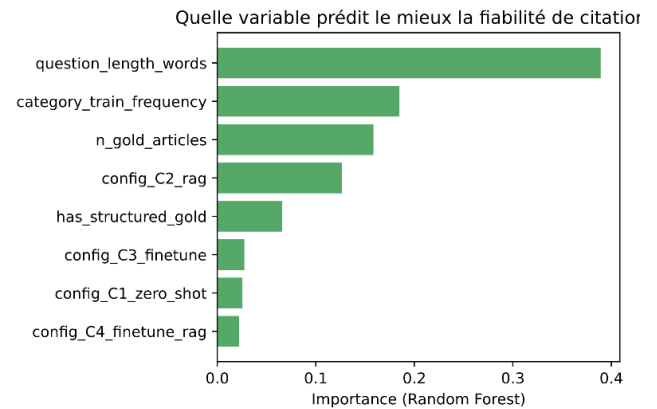


Figure 8. Feature importance for predicting citation exact match.

Feature importance shows that question length is the most important factor. Followed by the category's frequency in the training set and the number of gold articles. By contrast, the configuration used has a fairly weak influence. This means that the question's own characteristics have more impact on answer reliability than the choice between RAG and fine-tuning. This result thus echoes the earlier observation about the number of required articles.

By contrast, these same variables explain only about 13 % of the variation in the ROUGE-L score, whereas they allow much better prediction of citation exact match. This shows that finding the right article and producing an answer close to the expected one are two things that do not depend on exactly the same factors. It is therefore preferable to keep these two types of measures separate rather than combine them into a single score.

5.6 Fine-tuning ablations

To avoid drawing a conclusion from a single fine-tuning experiment, seven additional variants were tested by changing one factor at a time relative to the main configuration (table 9).

Table 9. Fine-tuning variants.

Variant	n	Train loss	Val. loss
Main ($r=32$)	580	0,834	0,679
Rank $r=8$	580	0,850	0,710
$n=190$	190	1,122	1,098
$n=380$	380	0,966	0,860
$n=786$	786	0,736	0,561
Attn + MLP	580	0,527	0,416
+ Synthetic	2580	0,940	0,516

Increasing the training set size steadily improves validation loss with no sign of saturation (figure 9), the point at 786 examples, the maximum available in BSARD, not having reached a plateau yet. Going from rank 8 to rank 32 (at equal number of examples) improves validation loss from 0.710 to 0.679, a real but modest gain for four times as many trained parameters. Extending the adapter to the feed-forward layers in addition to attention (Attn + MLP column) markedly lowers this loss, the largest absolute gain, but at the cost of a wider gap between training loss and validation loss, a sign of stronger overfitting. Adding 2000 automatically generated examples to the official training set gives the best validation loss among all the variants tested, an encouraging result that would deserve confirmation through a qualitative review of these synthetic examples before being fully adopted.

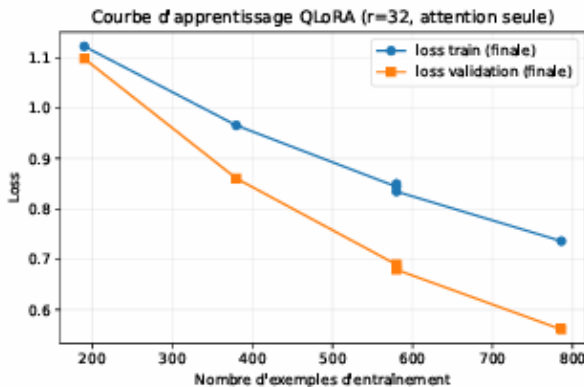


Figure 9. Validation loss by training set size.

The main configuration was also repeated across three random seeds, since a single run is not enough to draw a solid conclusion. The final training loss is stable across seeds, but the quality of the actually generated answers varies more. For the combined configuration, the mean ROUGE-L across the three seeds is lower than the one reported in table 4, which actually corresponds to the most favorable of the three seeds rather than to

a representative average. On ROUGE-L, the combined configuration only beats fine-tuning alone on two of the three seeds, which shows that its superiority, although real on average, is not systematic and should be qualified rather than presented as an established fact.

One final experiment tested whether an adapter trained on a single legal domain does better on that domain than a general-purpose adapter trained on everything. A specialist adapter, trained only on the 331 training questions touching the Code Civil (versus 580 for the generalist), reaches a training loss close to that of the main configuration, but its validation loss explodes, the largest overfitting gap among all the variants tested. On its own domain, this specialist is slightly better in ROUGE-L than the generalist but hallucinates markedly more. A limitation must be clearly stated here. The validation set used to measure this loss contains questions from all domains, not only the Code Civil, which mechanically disadvantages the specialist, evaluated on questions much broader than the ones it learned from. The specialist also has fewer training examples than the generalist, 331 versus 580. This experiment therefore mixes two different effects, that of restricting to a single domain and that of the amount of available data, and so cannot on its own tell whether specializing by domain truly helps. To cleanly separate these two effects, a validation set itself restricted to the Code Civil would be needed, which remains to be done.

5.7 Human evaluation and complementary automatic judgment

Every measure presented so far is automatic. To check whether it reflects a real human judgment, three annotators blindly compared the answers from RAG and from fine-tuning alone on 60 shared questions each, without knowing which answer came from which configuration.

Table 10. Inter-annotator agreement, Cohen’s Kappa by pair.

Pair	Kappa	Exact agreement	n
Bartosz vs Chaabane	0,389	56,7 %	60
Bartosz vs Arman	0,471	66,7 %	60
Chaabane vs Arman	0,331	53,3 %	60

Inter-annotator agreement, measured by Cohen’s Kappa (formulated in section 4.5), is moderate across the three pairs, between 0.33 and 0.47.

Table 11. Distribution of agreement over the 60 shared questions.

Agreement level	Share of questions
Unanimous (3 out of 3 votes)	45,0 %
Majority (2 out of 3 votes)	41,7 %
No agreement	13,3 %

Table 11 shows that the three annotators are unanimous in 45 % of cases and that at least majority agree-

ment is reached more than eight times out of ten, which remains reasonable for a subjective preference task.

Table 12. Preference by legal category in % of votes, three annotators combined.

Category	<i>n</i>	RAG	FT	Tie	Neither
Family	30	86,7	6,7	0,0	6,7
Money	30	76,7	3,3	10,0	10,0
Justice	30	70,0	3,3	16,7	10,0
Labor	18	72,2	0,0	11,1	16,7
Housing	30	66,7	6,7	6,7	20,0
Foreign nationals	30	60,0	20,0	16,7	3,3
Social protection	12	50,0	8,3	8,3	33,3

Table 12 confirms that RAG wins across all seven legal categories without exception, from 86.7 % preference in Family to 50 % in Social protection, the narrowest category with only 12 votes.

Table 13. Bradley-Terry ranking, on an Elo-like scale.

Configuration	Score
RAG (C2)	1368,7
Fine-tuning only (C3)	631,3

A Bradley-Terry model ranking, computed on the 161 usable votes out of the 180 collected (excluding the “neither” choice), clearly confirms this preference (table 13). This result stands in direct tension with the automatic comparison in section 5.2, which found no significant difference between RAG and fine-tuning on ROUGE-L. Rather than a contradiction to resolve, this gap concretely illustrates why a surface metric and a blind human judgment do not always point the same way, and justifies measuring them side by side instead of relying on a single type of signal.

In addition, a deterministic fidelity metric measures the overlap between the named entities and numbers extracted from the generated text and those from the reference text

$$\text{fidelity} = \frac{|PI_{\text{answer}} \cap PI_{\text{fragment}}|}{|PI_{\text{answer}}|},$$

with PI the set of points of interest (entities, numbers) extracted from a text. A relevance judgment from an independent LLM, Phi-3.5-mini-instruct, a 3.8-billion-parameter small language model released by Microsoft, chosen precisely because it is neither one of the compared systems nor the second base model tested, complements this measure, each answer being scored ten times at non-zero temperature and the scores averaged to avoid relying on a single unstable judgment, on a scale from 1 to 5.

Table 14. Fidelity to the reference text and LLM-judged relevance.

Configuration	Fidelity	Relevance /5
C1 zero-shot	0,054	5,00
C2 RAG	0,295	4,85
C3 fine-tune	0,298	4,73
C4 fine-tune + RAG	0,324	4,62

Table 14 shows an expected contrast on fidelity, very low for C1 with no context or adaptation at all (0.054), and markedly higher as soon as context or training comes into play, up to 0.324 for the C4 combination. Judged relevance, however, follows an inverse and more surprising trend, it is highest for C1 (5.00) and steadily decreases down to C4 (4.62). The judge’s prompt explicitly asks it not to assess factual accuracy, only whether the answer properly addresses what was asked, which mechanically explains this contrast. A zero-shot model answers the question directly and fluently, with no external context liable to lead it off topic, even when it fabricates the precise details. RAG, by contrast, can sometimes receive a poorly targeted passage from document retrieval, such as the example of the regulation on elite athlete status instead of a text on naturalization described in section 5.8, and the model then builds its answer from this wrong context rather than staying focused on the question. The retrieved context, which improves factual grounding on average, can therefore occasionally hurt perceived relevance, a real but markedly smaller effect than the citation-grounding gap described in section 5.3.

Table 14 also invites an additional hypothesis about the origin of the very marked human preference for RAG observed above. C2 and C3 scores are nearly identical here, 0.295 versus 0.298 in fidelity and 4.85 versus 4.73 in relevance, which is not enough to explain a human gap as sharp as 1368.7 versus 631.3 on the Bradley-Terry scale. A plausible explanation is that a human annotator also judges, at least in part, the form of the text rather than only its factual content, a better-structured answer with a clearer progression can be preferred even when its actual grounding is not better. None of the measures used here, ROUGE-L, fidelity, or judged relevance, explicitly captures the structure or readability of the produced text. The average answer length, a crude proxy for form, does not on its own support such a simple explanation, C3 produces on average the longest answers among the four configurations and yet clearly loses to C2, which is shorter, in the human arena, which at least rules out length alone as an explanatory factor. This form hypothesis remains to be tested directly, for instance by asking annotators to rate content and form separately, a distinction the current protocol does not make.

5.8 Qualitative error analysis

Beyond the numbers, directly reading the annotated answers reveals recurring error patterns that give substance to the statistics above. The most frequent is a legal role

reversal, an obligation attributed to the wrong party, often on pairs of questions that differ by a single word (for example “married” versus “legally cohabiting”), where the correct article differs in a subtle way that is easy to miss for both retrieval and generation. Fabricated statutory text also appears often. Since hallucination does not fully disappear even under oracle context (section 5.3), at least part of this phenomenon seems to be a general tendency of the base model rather than a mere consequence of poor retrieval. A smaller number of answers drift entirely away from the question asked, generally when a retrieved passage is close to the topic without really addressing it, which explains the slight degradation in judged relevance observed in section 5.7. Finally, a last pattern of repeated or cut-off text looks more like a decoding-configuration issue than a limitation of the model itself, since generation currently uses no explicit repetition penalty.

Two concrete examples illustrate these patterns, the first one quantified over the whole test set rather than on an isolated case. Article 1475 of the Code Civil, on legal cohabitation, appears in 31 answers out of 222 for fine-tuning alone, that is 14.0 %, versus only 2 for RAG, 2 for the RAG and fine-tuning combination, and none for zero-shot. Among these 31 answers, only two actually correspond to a question whose gold article is 1475, the remaining 29, that is 13.1 % of the whole test set for this configuration, cite this article even though it does not match the question asked. The model seems to have memorized a training example on this theme and to reproduce it almost verbatim as soon as it recognizes similar vocabulary, without really addressing the question asked. A good part of these citations are not counted as hallucinations by our automatic metric since the cited article really exists, which shows that a wrong but plausible citation largely escapes this measure. This behavior is specific to fine-tuning alone, it drops back down to RAG’s level as soon as RAG is added, which suggests that the retrieved context also helps pull the model out of this memorized pattern rather than merely providing extra information.

On a question about Belgian naturalization, zero-shot Mistral, for its part, invents an administrative body, the “Office fédéral de l’immigration et de l’intégration”, which does not exist in Belgium under that name, the competent body being the “Office des étrangers”. The corresponding acronym actually refers to a French administrative body, a concrete example of confusion between national administrative systems. On this same naturalization question, RAG gets it wrong in a different way, the search engine retrieves completely off-topic passages, notably a regulation on elite athlete status, and the resulting answer is shaped entirely by this irrelevant material, never once mentioning naturalization. Since generation depends directly on what retrieval provided, a retrieval-time error mechanically carries through to the final answer, a chain that helps explain why RAG gets particularly low scores on the Foreign nationals category, only 13 test questions, where a few retrieval errors like this one are enough to drag down the whole category’s

average.

5.9 Answers to the research questions and positioning

On the first question, can a model specialized through fine-tuning rival a general-purpose model backed by RAG, the answer is broadly yes on automatic metrics, C3 not being statistically distinguishable from C2 on ROUGE-L, but clearly no from a human standpoint, where RAG is very strongly preferred. This is exactly the kind of gap this work aimed to highlight by measuring both types of signal side by side.

On the second question, fine-tuning alone markedly improves the hallucination rate compared to all other configurations, which is a real and specific gain. Combined with RAG, it further improves mean ROUGE-L, but this superiority is confirmed neither by the significance tests nor systematically across the three tested seeds, the gain exists on average but is not guaranteed on a single run.

On the third question, the answer is clearly negative on the second part. The correct abstention rate on out-of-scope questions stays close to zero for all tested configurations, and this is the most serious problem in this work, more so than the choice between RAG and fine-tuning.

As in the reference study presented in section 3.3, we find that RAG is markedly preferred by human annotators over fine-tuning alone, with a comparable gap magnitude. Where our results diverge is on automatic metrics, that study found RAG systematically better than fine-tuning on judged relevance and fidelity, whereas we find no significant difference between RAG and fine-tuning on ROUGE-L, and even find fine-tuning markedly better on the hallucination rate. The most likely explanation, already discussed in section 3.3, is methodological, our fine-tuning is supervised on question-answer pairs whereas the other study’s was unsupervised continued pre-training.

6. Conclusion and future work

Several limitations should be kept in mind when reading these results. BSARD covers French-speaking Belgian law, not French law. The absence of a written gold answer forces the reconstruction of a reference by concatenating articles, which weighs on the interpretation of the surface similarity scores (section 4.1). Only two base model families, Mistral and Qwen, were tested, which leaves open the question of whether the observed patterns would generalize to much larger models or ones trained differently. And the inter-annotator agreement in the human arena, although honestly reported, remains moderate rather than strong.

This work compares RAG and QLoRA fine-tuning on a French legal question-answering assistant, with a protocol extended relative to a small-sample comparison, a complete test set, confidence intervals, paired tests, about ten additional ablations, retrieval-versus-

generation error decomposition, and a three-annotator human evaluation. RAG and fine-tuning each bring a clear and distinct gain over no adaptation, RAG mainly improves citation exact match, fine-tuning mainly improves caution against fabrication. Combining them further improves the result on average, without this gain being systematic from one run to another. The most serious problem uncovered is not a lack of quality on good questions, but the system’s near-total inability to recognize bad ones.

Taken together, these results suggest a practical recommendation rather than a perfect model. If the goal is strict source traceability, RAG is the most reliable option and the easiest to keep up to date. If the goal is to avoid outright fabrication, supervised fine-tuning brings a specific gain that RAG alone does not. Combining both improves the average result but adds variance and doubles the maintenance cost, which does not make it an obvious default choice in every deployment context.

Several directions remain open for future work. An adversarial stress test built specifically around plausible but nonexistent article references would help better characterize abstention ability. For the rarest categories such as Social protection, where the number of questions is too small to decide with confidence, a statistical power calculation would help determine whether the lack of a significant result reflects a true absence of effect or simply an insufficient sample. Further improving document retrieval itself also remains a central direction, since it is what bounds everything else, even the best variant tested in section 5.1 only ranks the correct article first in 15.1 % of cases, and no downstream writing quality can make up for a missing article from the start. But by far the most promising direction remains focused work on abstention ability, since it is the most serious weakness this work has revealed.

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Appendix A. Technical difficulties encountered

This appendix summarizes, in condensed form, the main technical difficulties encountered throughout the project.

- **Article identifier column type.** This BSARD field is a string rather than a list. Missing parsing silently drops Recall@k close to zero with no explicit error.
- **Heterogeneity of article identifiers.** About 32 % of BSARD article identifiers are not purely numeric (heterogeneous Belgian regional formats), and about 2.5 % carry a stray annotation suffix stuck on by the source scraping. Both issues silently skewed the citation metrics before being fixed with a generalized regular expression.
- **Lack of incremental saving.** Several long ablation jobs lost already-computed results due to a GPU memory overflow occurring before the final save. Generalized fix, save after each independent step rather than only at the end of the job.
- **Kaggle GPU session quota.** The limit of two simultaneous GPU sessions on the free account was exceeded several times with no explicit error message, notably because pushing a new version of a kernel does not automatically stop the previous session still running.
- **Duplicates in human arena votes.** One of the three annotators' vote files contained questions voted on twice, likely due to a page reload during the session. Fixed by systematically deduplicating by question identifier before any inter-annotator agreement computation.

Appendix B. Implementation details

Full QLoRA configuration. Rank $r = 32$, $\alpha = 32$, dropout 0.05, targets `q_proj`, `k_proj`, `v_proj`, `o_proj`, 4-bit NF4 quantization, three epochs, batch size 4 with gradient accumulation 4, learning rate 2×10^{-4} , base model `Mistral-7B-Instruct-v0.3`.

Embedding model and reranker. *multilingual-e5-large* for dense retrieval, *bge-reranker-v2-m3* for cross-encoder reranking, BM25Okapi for lexical retrieval, reciprocal rank fusion ($k = 60$) for the hybrid.

LLM judge. `Phi-3.5-mini-instruct`, chosen because it is absent from the four compared configurations and from the second base model tested, queried ten times per answer at non-zero temperature, scores averaged.